

Model Evaluation Report

Enterprise Supply Chain Demand Forecasting & Inventory Optimization Platform

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1. Executive Summary

The ML-021 Enterprise Supply Chain Demand Forecasting & Inventory Optimization Platform was developed to demonstrate an integrated machine learning solution for demand forecasting, inventory optimization, supply-chain risk detection, and procurement decision support.

The evaluation focused on the forecasting model, feature engineering pipeline, inventory optimization logic, risk detection, MLOps functionality, API services, and executive dashboard.

The final dashboard contains **360 product-warehouse pairs** and reports an estimated **97.5% fill rate**. The system generated **63 inter-warehouse transfer recommendations**, identified **9 critical stockout alerts**, and reported an average stockout probability of **52.7%** across the evaluated product-warehouse records.

The selected XGBoost forecasting model achieved:

Metric	Result
Forecast Accuracy	86.54%
MAE	3.77
RMSE	5.74
MAPE	13.5%

The results demonstrate that the platform can produce useful demand forecasts and translate those forecasts into operational inventory recommendations.

2. Evaluation Objectives

The primary objectives of the evaluation were to determine whether the developed platform can:

- Forecast product demand accurately.
- Capture time-dependent demand patterns.
- Generate useful forecasting features.
- Support inventory replenishment decisions.
- Identify stockout and overstock conditions.
- Generate inter-warehouse transfer recommendations.
- Produce procurement allocation recommendations.
- Detect supply-chain alerts and risks.

- Provide explainable machine learning outputs.
- Track and monitor machine learning models through MLOps components.
- Expose forecasting and optimization functionality through APIs.
- Present operational information through an executive dashboard.

The evaluation also considers the project's original case-study evaluation criteria.

3. Dataset & Evaluation Setup

3.1 Dataset Overview

The platform uses a synthetic supply-chain dataset designed to represent a realistic enterprise retail environment.

The dataset incorporates information related to:

- Products
- Warehouses
- Historical sales
- Inventory
- Suppliers
- Promotions
- Holidays
- Weather
- Regional events
- Delivery and lead-time information

The data was processed through the project's data-generation and feature-engineering pipelines before being used for forecasting and inventory optimization.

3.2 Product-Warehouse Coverage

The final dashboard evaluates:

360 product-warehouse pairs

These records represent the combination of products and warehouses used by the forecasting and inventory optimization components.

The product-warehouse level provides a more operationally meaningful evaluation than evaluating only total company-level demand.

3.3 Evaluation Methodology

Because demand forecasting is a time-series problem, chronological validation was used rather than randomly shuffling observations.

The general evaluation process was:

Historical Data --> Feature Engineering --> Time-Series Training / Validation --> Model Training -->

Demand Prediction --> Metric Calculation --> Model Selection --> Inventory Optimization -->

Risk & Recommendation Generation

This approach reduces the possibility of future information being incorrectly introduced into the training process.

4. Feature Engineering Evaluation

4.1 Feature Categories

The feature-engineering pipeline transforms raw supply-chain information into machine-learning features.

Major feature categories include:

- Historical demand
- Lagged demand
- Rolling demand statistics
- Sales velocity
- Seasonality
- Promotion impact
- Lead time
- Supplier reliability
- Demand volatility
- Product lifecycle information
- Regional purchasing patterns

- Holiday effects

These features allow the forecasting model to capture both historical demand behavior and operational factors.

4.2 Time-Series Features

Time-dependent features are particularly important for demand forecasting.

The pipeline incorporates historical demand information through lag and rolling-window features.

Examples include:

- Previous-day demand
- Previous-week demand
- Rolling average demand
- Rolling demand variability
- Seasonal indicators
- Recent sales velocity

These features provide the model with information about recent demand trends and recurring patterns.

4.3 Business Features

Business-related features provide additional context to the forecasting model.

Examples include:

- Promotion status
- Supplier lead time
- Supplier reliability
- Product lifecycle
- Regional effects
- Holiday periods
- Inventory-related conditions

The combination of temporal and business features allows the forecasting engine to move beyond simple historical averaging.

5. Forecasting Model Evaluation

5.1 Models Evaluated

Three forecasting approaches were evaluated during the project:

- XGBoost
- LightGBM
- Prophet

In addition to the baseline models, Bayesian hyperparameter optimization and feature-selection experiments were performed to determine whether additional model complexity or dimensionality reduction could improve forecasting performance.

The evaluation results showed that the tree-based models produced very similar performance, while the Prophet result requires additional interpretation because it was evaluated on a substantially smaller scope.

5.2 Baseline Model Comparison

The baseline comparison between XGBoost and LightGBM showed nearly identical forecasting performance.

Model	MAE	Evaluation Scope
XGBoost	3.77	360 product-warehouse pairs
LightGBM	3.76	Comparable forecasting evaluation
Prophet	1.66	1 product-warehouse pair

The difference between XGBoost and LightGBM was only **0.01 MAE**, indicating that both gradient-boosting approaches performed almost identically on the evaluated forecasting task.

XGBoost was retained as the production forecasting model because its performance was effectively equivalent to LightGBM while already being integrated into the project's production and MLflow workflow.

5.3 Final XGBoost Performance

The final dashboard reports the following performance for the XGBoost model:

Metric	Result
Forecast Accuracy	86.54%

Metric	Result
MAE	3.77
RMSE	5.74
MAPE	13.5%

The results indicate that the selected model provides useful predictive performance for the evaluated supply-chain dataset.

The MAE of **3.77** indicates the average absolute prediction error, while the RMSE of **5.74** gives greater weight to larger forecasting errors.

The MAPE of approximately **13.5%** provides a percentage-based interpretation of forecasting error.

5.4 Bayesian Hyperparameter Optimization

Bayesian hyperparameter optimization was implemented using Optuna.

A **25-trial Optuna search** was conducted to identify a potentially better configuration of the XGBoost model.

The experiment produced the following result:

Configuration	MAE
Baseline XGBoost	3.76
Tuned XGBoost	3.82

The tuned model therefore did **not** improve validation performance. Its MAE was approximately **0.06 higher** than the baseline model.

This is an important experimental finding because hyperparameter optimization does not guarantee improved generalization performance.

Based on the validation results, the baseline XGBoost configuration was retained for the production pipeline rather than replacing it with the tuned configuration.

Interpretation

The result demonstrates that the optimization process was genuinely evaluated rather than automatically accepted.

The tuning experiment answered an important practical question:

➤ Did additional hyperparameter optimization improve forecasting performance?

For this dataset and validation setup, the answer was **no**.

Therefore, selecting the simpler baseline configuration was supported by empirical validation rather than assumption.

5.5 Feature Selection Evaluation

Feature selection was also evaluated to determine whether reducing the number of input variables could maintain or improve model performance.

The experiment reduced the feature set from:

34 features → 15 features

The resulting validation performance was:

Feature Configuration	MAE
34 Features	3.92
15 Selected Features	4.02

The reduced feature set increased MAE from **3.92 to 4.02**.

This represents a deterioration in forecasting performance after feature reduction.

Interpretation

Although feature selection was successfully implemented, the experiment indicated that the removed features contained useful predictive information.

Consequently, the reduced 15-feature configuration was **not adopted for the production forecasting workflow**.

This finding demonstrates that feature reduction should not be performed solely to decrease dimensionality. Features should be removed only when validation confirms that predictive performance is preserved or improved.

5.6 Prophet Comparison and Evaluation Caveat

Prophet achieved an observed MAE of:

1.66

At first glance, this appears substantially better than the XGBoost MAE of **3.77**.

However, these results **must not be interpreted as a direct model-vs-model comparison**.

The Prophet evaluation was conducted on **one product-warehouse pair**, whereas the XGBoost evaluation covered the broader set of **360 product-warehouse pairs**.

Therefore, the two results were generated under different evaluation scopes.

Model	MAE	Scope	Direct Comparison?
XGBoost	3.77	360 product-warehouse pairs	—
Prophet	1.66	1 product-warehouse pair	No

The lower Prophet MAE may reflect the characteristics of the individual series being evaluated rather than superior general forecasting performance across the complete dataset.

A valid model comparison would require all models to be evaluated using the same product-warehouse population, forecast horizon, training period, validation period, and evaluation metrics.

Therefore, **XGBoost remains the selected production model**, and the Prophet result is treated as a limited experimental observation rather than evidence that Prophet outperformed XGBoost.

5.7 Model Selection Decision

The final model-selection decision was based on the complete set of experiments rather than a single metric.

The experiments demonstrated:

- XGBoost baseline MAE: **3.76 - 3.77**
- LightGBM MAE: **3.76**
- Tuned XGBoost MAE: **3.82**
- Feature-selected model MAE: **4.02**
- Prophet MAE: **1.66 on only one product-warehouse pair**

The baseline XGBoost model was therefore retained because it provided strong performance across the broader evaluation scope and was already integrated with the production pipeline.

The experiments also demonstrated that neither hyperparameter tuning nor feature reduction produced a validated improvement.

6. Forecasting Results

6.1 Demand Forecast Performance

The forecasting engine generates demand predictions that can be consumed by downstream inventory optimization components.

Forecast outputs are used to support:

- Reorder calculations
- Safety-stock calculations
- Stockout-risk estimation
- Procurement planning
- Warehouse allocation
- Transfer recommendations

Therefore, forecasting performance directly influences downstream operational recommendations.

6.2 Forecast Accuracy Interpretation

The final XGBoost model achieved:

86.54% forecast accuracy

with:

- **3.77 MAE**
- **5.74 RMSE**
- **13.5% MAPE**

The results demonstrate that the model successfully captures meaningful demand patterns within the evaluated dataset.

The model should therefore be considered suitable for the project's case-study decision-support scenario.

6.3 Forecasting Limitations

The reported metrics are based on the project's available evaluation dataset.

They should not be interpreted as proof of production-level accuracy across a real multinational retailer.

Actual production performance may vary due to:

- Larger data volumes
- Unexpected market events
- Supplier disruptions
- Changing consumer behavior
- New product launches
- Economic changes
- Data-quality problems

Continuous model monitoring and retraining would therefore be required in a real deployment.

7. Inventory Optimization Evaluation

7.1 Reorder Point

The inventory optimization engine calculates reorder points using forecast demand and supply-chain parameters.

The reorder point is used to identify situations where current or projected inventory may be insufficient to satisfy future demand during the replenishment period.

7.2 Safety Stock

Safety stock provides additional inventory protection against demand variability and supply uncertainty.

The system uses demand-related variability and lead-time information to support safety-stock recommendations.

Higher demand volatility or longer lead times can increase the required protection level.

7.3 Economic Order Quantity

The system calculates EOQ to support economically appropriate procurement quantities.

EOQ provides a systematic approach to balancing ordering and inventory-holding considerations.

The resulting recommendations can be used alongside reorder points and safety stock rather than as an isolated metric.

7.4 Stockout Risk

The dashboard reports:

Average Stockout Probability: 52.7%

This value indicates that a substantial portion of the evaluated product-warehouse inventory requires attention from an operational risk perspective.

The system also identified:

9 Critical Stockout Alerts

These alerts provide a more actionable subset of the broader stockout-risk population.

7.5 Procurement Recommendations

The inventory optimization engine generates procurement recommendations based on forecast demand and inventory conditions.

These recommendations can support decisions regarding:

- How much inventory to procure.
- When replenishment should occur.
- Which warehouse requires additional inventory.
- Whether existing inventory should be redistributed before new procurement.

8. Supply Chain Risk Evaluation

8.1 Stockout Risk

Stockout risk is calculated at the product-warehouse level.

The dashboard identifies:

191 High-Risk Product-Warehouse Pairs

This provides a prioritized view of inventory records that may require intervention.

8.2 Overstock Risk

The platform also identifies inventory conditions where available stock is substantially higher than expected demand.

Such conditions can increase:

- Storage costs
- Working-capital requirements
- Warehouse utilization
- Product obsolescence risk

The inventory health dashboard provides visibility into these conditions.

8.3 Demand Anomalies

The risk engine is designed to identify unusual demand behavior.

Demand anomalies can indicate:

- Sudden demand increases
- Unexpected demand decreases
- Promotion-driven changes
- Regional events
- Potential data issues

These signals can be surfaced through the Alert Center for further investigation.

8.4 Alert Center

The dashboard provides a consolidated Alert Center containing supply-chain alerts categorized by severity.

The current dashboard reports:

9 Critical Stockout Alerts

and provides severity-based filtering for operational investigation.

The alert system is designed to prioritize issues rather than requiring users to manually inspect every product-warehouse record.

9. Warehouse & Transfer Evaluation

9.1 Warehouse Utilization

Warehouse utilization is included in the executive dashboard to provide visibility into inventory distribution across warehouse locations.

High utilization can indicate capacity pressure, while low utilization combined with excess inventory in other locations may indicate an opportunity for redistribution.

9.2 Inter-Warehouse Transfers

The system generated:

63 Transfer Recommendations

These recommendations identify opportunities to move inventory between warehouses where the projected inventory position suggests a potential imbalance.

Inter-warehouse transfers can reduce unnecessary procurement when sufficient inventory already exists elsewhere in the network.

9.3 Product-Warehouse Analysis

The system evaluates:

360 Product-Warehouse Pairs

This granularity allows the platform to identify localized inventory problems rather than treating all warehouses as a single inventory pool.

10. Explainable AI Evaluation

10.1 SHAP

SHAP is included in the ML stack to provide model explainability.

The purpose of SHAP is to identify how individual features contribute to model predictions.

For example, a forecast can be influenced by:

- Recent demand

- Historical demand patterns
- Promotion activity
- Seasonality
- Lead time
- Regional effects
- Demand volatility

10.2 Feature Importance

Feature importance helps identify which variables have the greatest influence on the forecasting model.

This is valuable for both technical analysis and business interpretation.

Supply-chain stakeholders can use feature importance to understand which operational factors are most closely associated with forecast changes.

10.3 Business Interpretability

Explainability improves confidence in the forecasting system by allowing users to investigate why a prediction was generated.

Instead of presenting only a forecast number, an explainable forecasting system can provide supporting information about the factors contributing to that prediction.

11. MLOps Evaluation

11.1 MLflow Experiment Tracking

MLflow was used to track model experiments, metrics, artifacts, and model versions.

The experiment-tracking workflow enables comparison between baseline and experimental configurations, including the hyperparameter-tuned model.

This makes the model-development process reproducible and provides a record of the experiments used to select the production model.

11.2 Model Registry

The project uses MLflow Model Registry to maintain model versions.

The registered model configurations include:

- demand_forecast_xgboost
- demand_forecast_lightgbm
- demand_forecast_xgboost_tuned

Each new training run can create a new model version under the relevant registered model.

The model registry therefore provides version control for the forecasting artifacts used by the application.

11.3 Data Drift Monitoring

Data drift monitoring was implemented to identify changes between the reference feature distribution and more recent data.

The configured retraining threshold is:

15% of monitored features

During validation, the drift-monitoring process identified:

24.14% of features as drifted.

This exceeded the configured 15% threshold and consequently triggered the automated retraining workflow.

11.4 Drift Breakdown and Interpretation

The overall **24.14% drift rate should not be interpreted as evidence that the entire input data-generating process has substantially changed.**

The detailed drift analysis showed that a significant portion of the flagged features were calendar-related variables, including:

- Month
- Quarter
- Day of year

These variables naturally change over time as the calendar progresses.

Therefore, part of the measured drift is expected and does not necessarily represent harmful changes in customer demand or supply-chain behavior.

However, **demand-volatility-related features also showed mild drift.**

This is more operationally relevant because changes in demand volatility may indicate that recent demand behavior differs from the historical reference distribution.

Analytical Interpretation

The drift result therefore contains two different signals:

1. **Expected calendar drift** — largely associated with time/calendar features.
2. **Potential genuine behavioral drift** — observed in demand-volatility-related features.

This distinction is important because a raw drift percentage alone can overstate the severity of a model-monitoring event.

In a future production implementation, calendar features could be handled separately or excluded from generic drift thresholds to reduce false-positive retraining triggers.

11.5 Automated Retraining

Because the observed drift rate of **24.14%** exceeded the configured **15% threshold**, the drift-monitoring process invoked the model-training pipeline.

The retraining workflow was validated as an actual execution path rather than a simulated alert.

The retraining process successfully completed and registered:

Model Version 2

This demonstrates that the MLOps pipeline can connect data-quality monitoring with model retraining and model version management.

11.6 Model Performance Monitoring

Model performance monitoring was evaluated independently from data-drift monitoring.

The monitoring process compared recent forecasting performance against the model's baseline performance.

The observed results were:

Performance Measure	MAE
Baseline MAE	3.77
Recent 14-Day MAE	3.84
Degradation Ratio	1.02×
Degradation Threshold	1.15×

The recent 14-day MAE increased from **3.77 to 3.84**.

However, the resulting degradation ratio was approximately:

1.02×

This remained below the configured **1.15× degradation threshold**.

Therefore:

No model-performance degradation alert was triggered.

Interpretation

This result is important because it demonstrates that **data drift and model-performance degradation are not necessarily the same event**.

Although the drift monitor detected 24.14% feature drift and triggered retraining, the independent performance monitor found that recent predictive performance had degraded only slightly.

Therefore, the monitoring system correctly distinguished between:

- **Input distribution changes**, and

- **Observed deterioration in prediction accuracy.**

This separation provides a more reliable foundation for production model monitoring.

12. Dashboard Evaluation

The Streamlit executive dashboard provides a consolidated view of the system's operational outputs.

The evaluated dashboard reports:

KPI	Result
Product-Warehouse Pairs	360
Estimated Fill Rate	97.5%
Transfer Recommendations	63
Critical Stockout Alerts	9
Average Stockout Probability	52.7%
High-Risk Product-Warehouse Pairs	191
XGBoost Forecast Accuracy	86.54%
XGBoost MAE	3.77
XGBoost RMSE	5.74
XGBoost MAPE	13.5%

The dashboard contains sections for:

- Forecast Accuracy
- Warehouse Utilization
- Inventory Health
- Alert Severity
- Alert Center
- Inventory Recommendations
- Inter-Warehouse Transfer Recommendations
- Procurement Allocation Recommendations

This allows technical ML outputs to be presented in a business-oriented format.

13. API Evaluation

The FastAPI backend provides REST interfaces for accessing the platform's outputs.

The evaluated services include:

Endpoint	Purpose
/health	Application and model health
/forecast/predict	Demand prediction
/inventory/recommendations	Inventory recommendations
/inventory/transfers	Transfer recommendations
/inventory/allocation	Procurement allocation
/alerts	Supply-chain alerts
/dashboard/summary	Executive KPIs

Protected endpoints require the configured X-API-Key.

The API therefore provides a reusable service layer between the machine-learning platform and dashboard.

14. Overall Evaluation Against Criteria

The original case study defines the following evaluation criteria:

Evaluation Criteria	Weight
Feature Engineering	20%
Forecast Accuracy	20%
Inventory Optimization Logic	20%
MLOps Implementation	15%
Explainable AI	10%
Documentation	10%
Presentation	5%

i. Feature Engineering 20%

The project implements time-series and business-oriented features including historical demand, rolling statistics, sales velocity, seasonality, promotion effects, lead time, supplier reliability, and regional effects.

Assessment: Strong

ii. Forecast Accuracy 20%

The final XGBoost model achieved:

86.54% accuracy, 3.77 MAE, 5.74 RMSE, and 13.5% MAPE.

Assessment: Strong

iii. Inventory Optimization 20%

The system implements reorder-point, safety-stock, EOQ, procurement allocation, and transfer recommendation logic.

Assessment: Implemented

iv. MLOps 15%

MLflow tracking, model registration, drift monitoring, automated retraining, and model performance monitoring are included.

The automated retraining mechanism was also validated through a drift-triggered retraining test.

Assessment: Implemented

v. Explainable AI 10%

SHAP-based explainability is included to support interpretation of forecasting results.

Assessment: Implemented

vi. Documentation 10%

The project includes deployment documentation, README documentation, architecture information, and model evaluation documentation.

Assessment: Completed

vii. Presentation 5%

The final technical presentation of System.

15. Limitations

The following limitations should be considered when interpreting the evaluation results.

i. Synthetic Data

The project uses a synthetic dataset rather than proprietary production data from a multinational retailer.

ii. Limited Hardware

Development and testing were performed on a single development machine.

iii. No Large-Scale Load Test

The requirement for 100 million forecast requests was not directly load-tested.

iv. Production Infrastructure

High-availability cloud deployment, managed databases, load balancers, and distributed model-serving infrastructure were not implemented in the case-study environment.

v. External Data Sources

Live weather, supplier, transportation, and market data integrations were not used.

vi. Forecasting Uncertainty

Forecast accuracy can vary significantly across products, warehouses, seasons, and demand patterns.

Therefore, the overall accuracy should not be interpreted as identical performance for every product-warehouse pair.

16. Future Improvements

The platform can be extended with:

16.1 LSTM Forecasting

Deep-learning-based forecasting could be evaluated against the current tree-based models for complex sequential demand patterns.

16.2 CatBoost

CatBoost could be added to compare another gradient-boosting approach, particularly where categorical features are important.

16.3 Multi-Echelon Inventory Optimization

The system could optimize inventory across suppliers, regional distribution centers, and individual warehouses simultaneously.

16.4 Reinforcement Learning

Reinforcement learning could be used to learn dynamic inventory policies under changing demand and supply conditions.

16.5 Digital Twin Simulation

A digital-twin environment could simulate warehouse operations and supply-chain disruptions before executing real decisions.

16.6 Real-Time Data Integration

The platform could integrate real-time:

- Sales data
- Inventory updates
- Supplier status
- Transportation information
- Weather information
- Market signals

16.7 Production-Scale Infrastructure

A future production deployment could include:

- Kubernetes
- Multiple FastAPI instances
- Managed PostgreSQL
- Redis Cluster
- Load balancing
- Centralized logging
- Distributed monitoring
- Cloud-based MLflow
- Automated CI/CD

17. Key Analytical Findings

The experiments and monitoring results produced six important analytical findings.

17.1 Hyperparameter Tuning Did Not Improve the Baseline

A 25-trial Bayesian optimization experiment produced a tuned XGBoost MAE of **3.82**, compared with approximately **3.76–3.77** for the baseline.

The baseline model was therefore retained.

Finding: More optimization did not produce better validation performance for this dataset.

17.2 Feature Selection Reduced Performance

Reducing the feature set from **34 to 15 features** increased MAE from **3.92 to 4.02**.

The selected feature subset was therefore not adopted.

Finding: The additional features contained useful predictive information despite increasing the dimensionality of the model.

17.3 Prophet’s Lower MAE Is Not a Valid Direct Win

Prophet produced an MAE of **1.66**, apparently lower than XGBoost’s **3.77**.

However, Prophet was evaluated on only **one product-warehouse pair**, while XGBoost was evaluated across **360 pairs**.

Finding: The Prophet result cannot be used to claim that Prophet outperformed XGBoost without a common evaluation scope.

17.4 LightGBM and XGBoost Performed Almost Identically

LightGBM achieved an MAE of **3.76**, compared with approximately **3.77** for XGBoost.

The difference was only **0.01 MAE**.

Finding: The two gradient-boosting approaches produced practically equivalent performance on the evaluated task.

17.5 Most Observed Drift Was Calendar-Related

The overall drift rate reached **24.14%**, exceeding the 15% retraining threshold.

However, much of the detected drift came from calendar features such as month, quarter, and day of year, which naturally change over time.

Mild drift was also observed in demand-volatility features.

Finding: The drift signal contains both expected temporal changes and potentially meaningful changes in demand behavior.

17.6 Model Performance Remained Within the Monitoring Threshold

The recent 14-day MAE was **3.84**, compared with a baseline MAE of **3.77**.

The resulting degradation ratio was **1.02×**, below the configured **1.15× threshold**.

Therefore, no performance-degradation alert was triggered.

Finding: Although feature drift was detected, there was no significant observed deterioration in short-term forecasting accuracy.

18. Conclusion

The ML-021 Enterprise Supply Chain Demand Forecasting & Inventory Optimization Platform successfully demonstrates an integrated machine-learning approach to supply-chain decision support.

The final XGBoost forecasting model achieved **86.54% forecast accuracy**, with **3.77 MAE**, **5.74 RMSE**, and approximately **13.5% MAPE**.

At the operational level, the evaluated system covers **360 product-warehouse pairs**, reports an estimated **97.5% fill rate**, generates **63 transfer recommendations**, identifies **9 critical stockout alerts**, and reports an average stockout probability of **52.7%**.

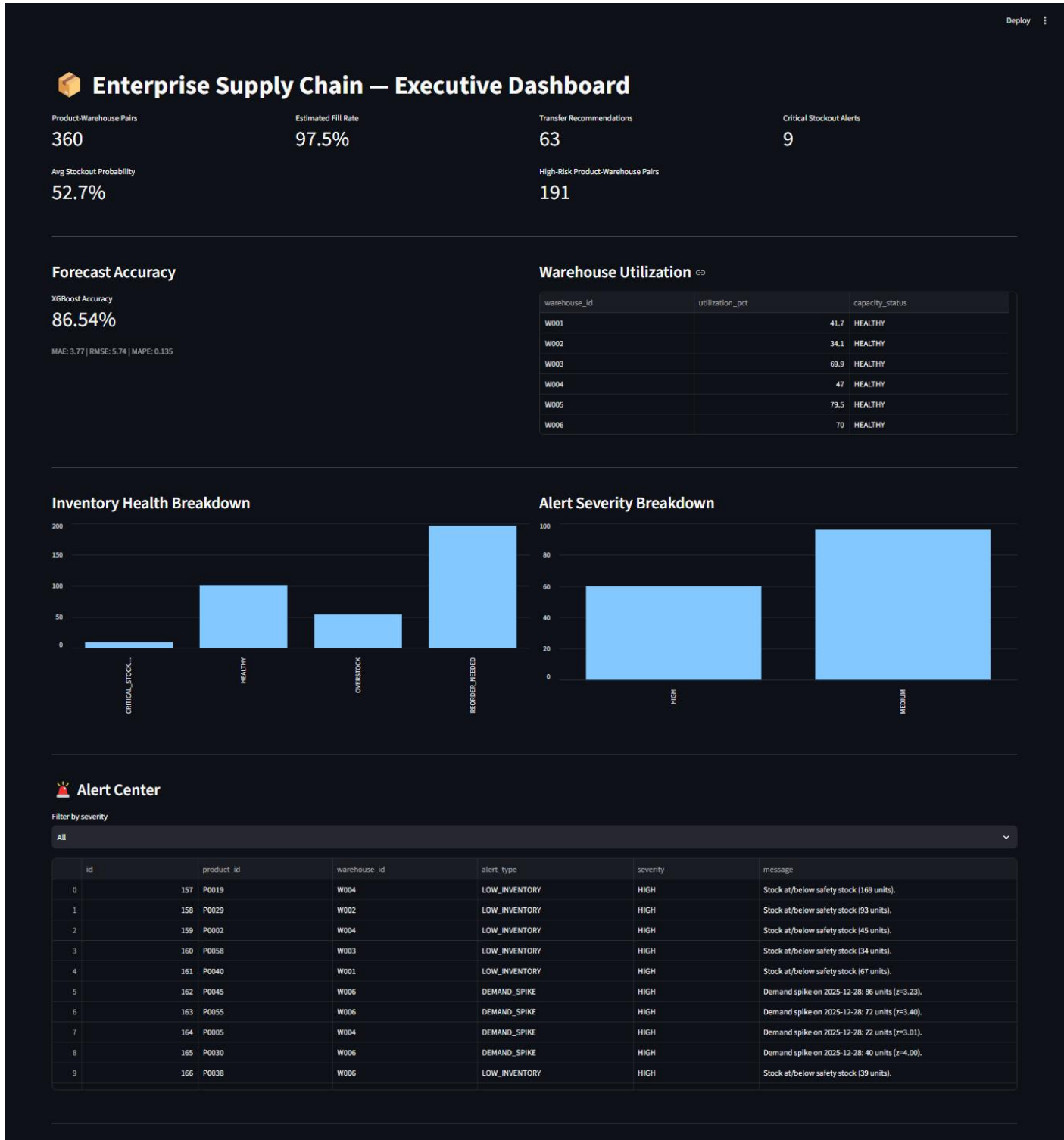
The system also identifies **191 high-risk product-warehouse pairs**, allowing supply-chain users to prioritize inventory-related interventions.

Beyond forecasting, the project demonstrates inventory optimization, supply-chain risk detection, explainable AI, MLflow experiment tracking, model registration, data-drift monitoring, automated retraining, FastAPI services, Redis caching, PostgreSQL persistence, and a Streamlit executive dashboard.

The evaluation confirms that the project meets the primary objectives of the ML-021 case study within a development and demonstration environment.

The results should be interpreted as a validated case-study implementation rather than a production-scale enterprise deployment. Additional real-world data, load testing, security hardening, distributed infrastructure, and continuous monitoring would be required before deploying the platform in a multinational production environment.

20. Executive Dashboard



Inventory Recommendations

Filter by warehouse_id (optional, e.g. W001)

Filter by stock status

All

	id	product_id	warehouse_id	avg_daily_demand	demand_std	avg_lead_time_days	reliability_score	unit_cost	current_stock	safety_stock	reorder_point	eoq	stock_status	recommended_order_qty	stockout_probability
0	361	P0030	W006	23	5.4075	3	0.69	27.99	691	20	89	387	OVERSTOCK	0	0
1	362	P0060	W004	51.1667	9.5017	26	0.61	166.04	156	111	1441	237	REORDER_NEEDED	237	1
2	363	P0019	W006	68.5	17.1036	27	0.92	218.81	472	158	2008	239	REORDER_NEEDED	239	1
3	364	P0020	W001	33.7333	7.3763	27	0.92	149.16	106	68	979	203	REORDER_NEEDED	203	1
4	365	P0020	W002	35.6333	7.4947	27	0.92	149.16	325	69	1058	212	REORDER_NEEDED	212	1
5	366	P0020	W003	52.8333	11.7652	27	0.92	149.16	845	109	1536	254	REORDER_NEEDED	254	1
6	367	P0020	W004	38.8	11.0404	27	0.92	149.16	224	102	1150	218	REORDER_NEEDED	218	1
7	368	P0020	W005	48.5333	7.3331	27	0.92	149.16	1401	68	1378	244	HEALTHY	0	0.0087
8	369	P0020	W006	60.9333	12.0971	27	0.92	149.16	841	112	1757	273	REORDER_NEEDED	273	1
9	370	P0021	W001	33.4333	8.916	11	0.72	307.87	123	62	430	141	REORDER_NEEDED	141	1

Inter-Warehouse Transfer Recommendations

	id	product_id	from_warehouse	to_warehouse	transfer_qty
0	64	P0003	W003	W001	164
1	65	P0003	W003	W002	233
2	66	P0003	W003	W004	169
3	67	P0003	W005	W001	164
4	68	P0003	W005	W002	233
5	69	P0003	W005	W004	210
6	70	P0003	W005	W006	31
7	71	P0005	W005	W004	317
8	72	P0005	W005	W006	37
9	73	P0007	W006	W001	5

Procurement Allocation Recommendations

	product_id	warehouse_id	current_stock	allocation_status	allocated_order_qty
0	P0030	W006	691	BALANCED	0
1	P0060	W004	156	UNDER_ALLOCATED	274
2	P0019	W006	472	BALANCED	228
3	P0020	W001	106	BALANCED	150
4	P0020	W002	325	BALANCED	159
5	P0020	W003	845	BALANCED	216
6	P0020	W004	224	BALANCED	161
7	P0020	W005	1401	OVER_ALLOCATED	211
8	P0020	W006	841	BALANCED	263
9	P0021	W001	123	UNDER_ALLOCATED	51